DEVOPS

LAB 06

Set Up Nexus Repository and Configure User Access

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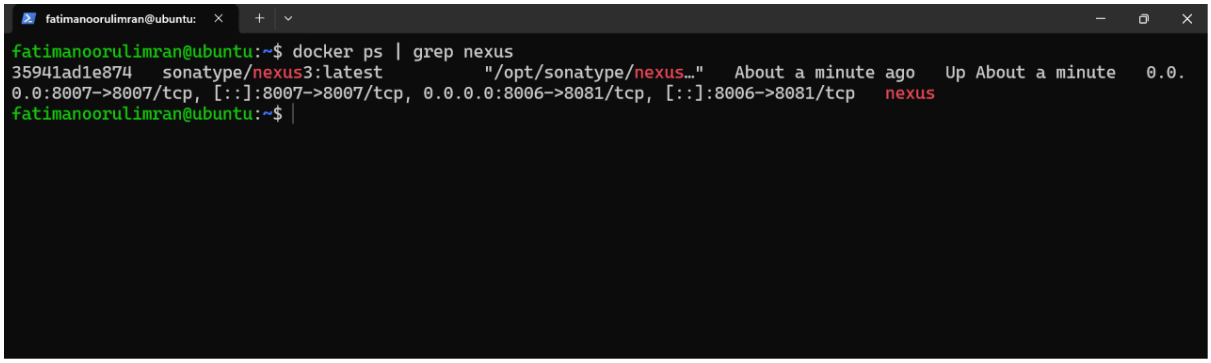
Registration Number: 2023-BSCS-33

Course Code: BCS-649

Section: A

GitHub Username: fatimanoorulimran

Task 1: Run Nexus Repository and Complete the Initial Setup



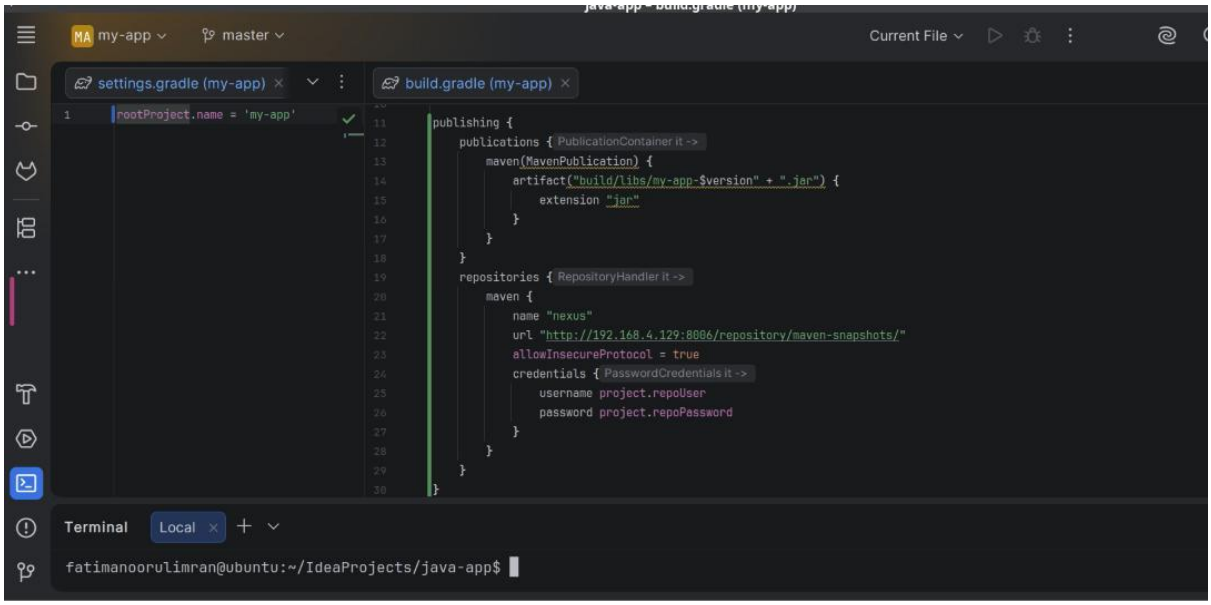
docker ps | grep nexus shows the nexus container running with ports 8006 and 8007 published.

Task 2: Create a Role and User and Verify Access

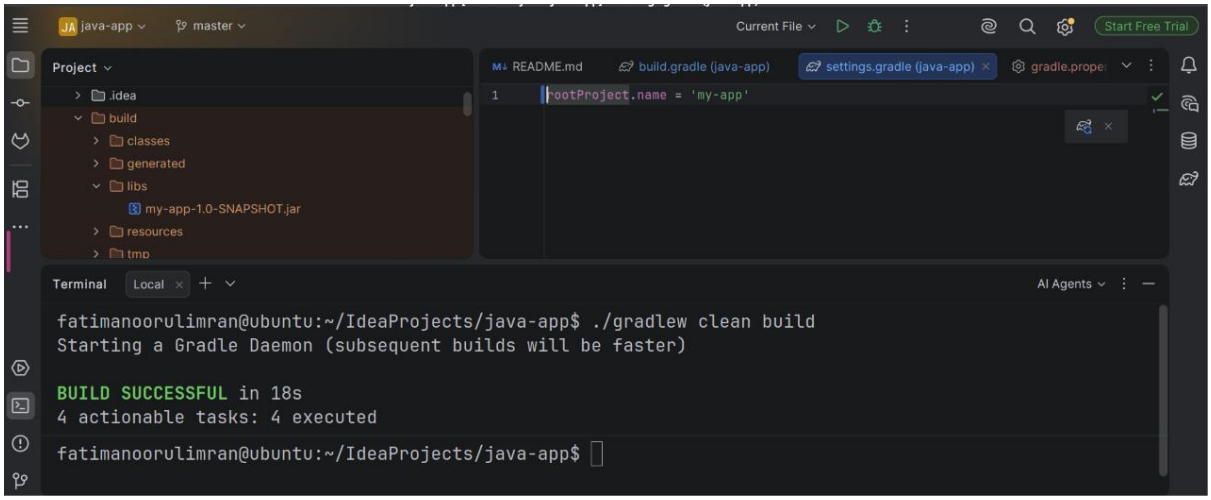


The nexus user authenticated to the Status API and received HTTP/1.1 200 OK.

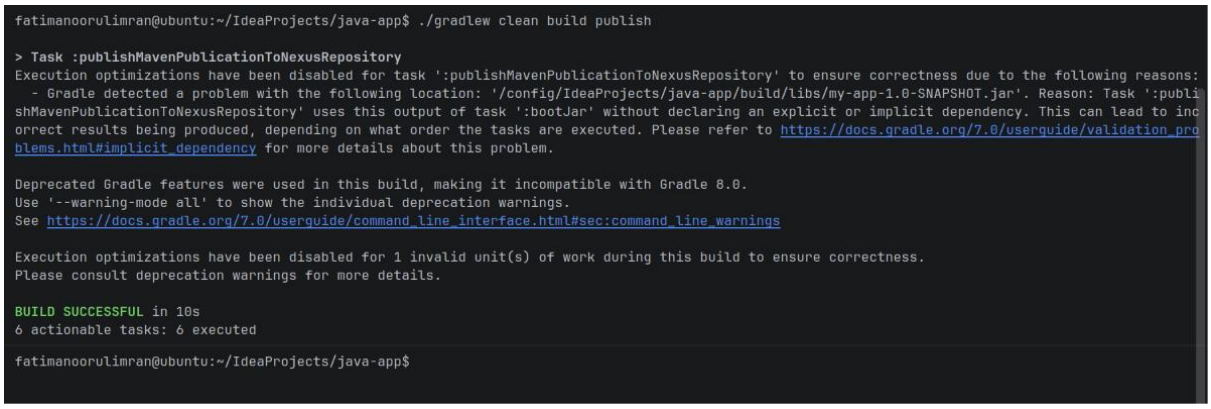
Task 3: Build and Publish the Gradle Application to Nexus



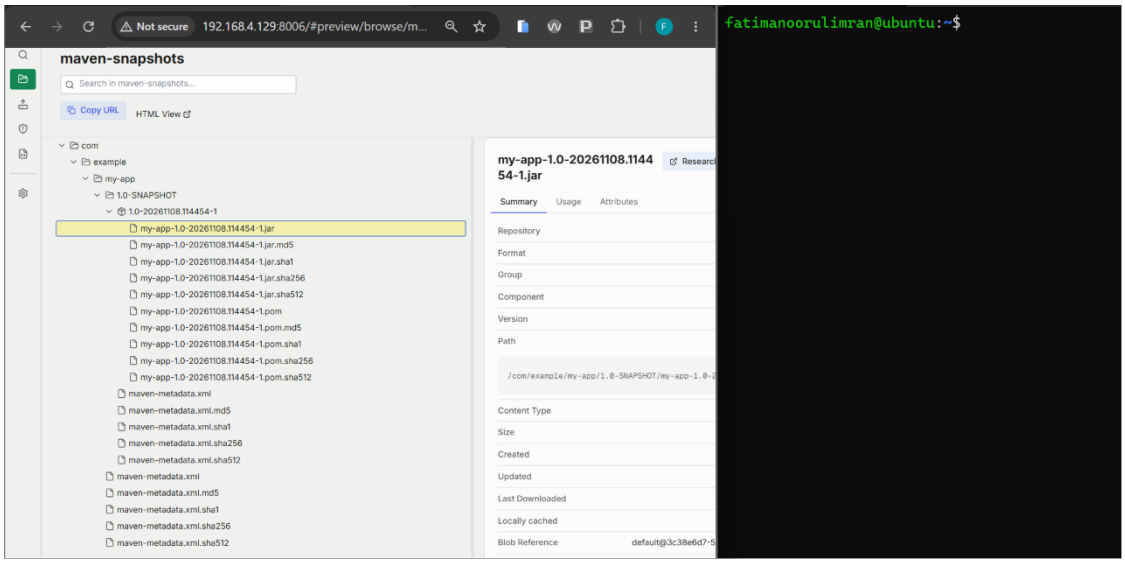
build.gradle publishing configuration and settings.gradle with rootProject.name set to my-app.



./gradlew clean build completed with BUILD SUCCESSFUL and generated my-app-1.0-SNAPSHOT.jar.

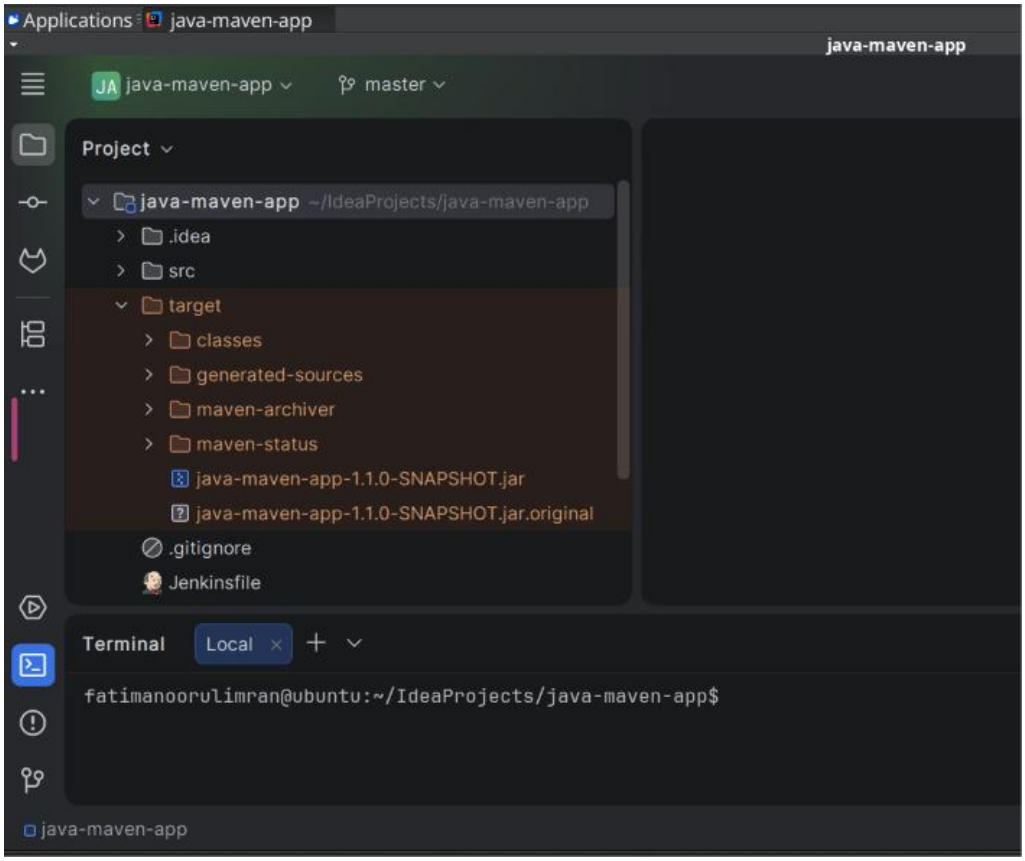


./gradlew clean build publish ran the Nexus publish task and finished with BUILD SUCCESSFUL.

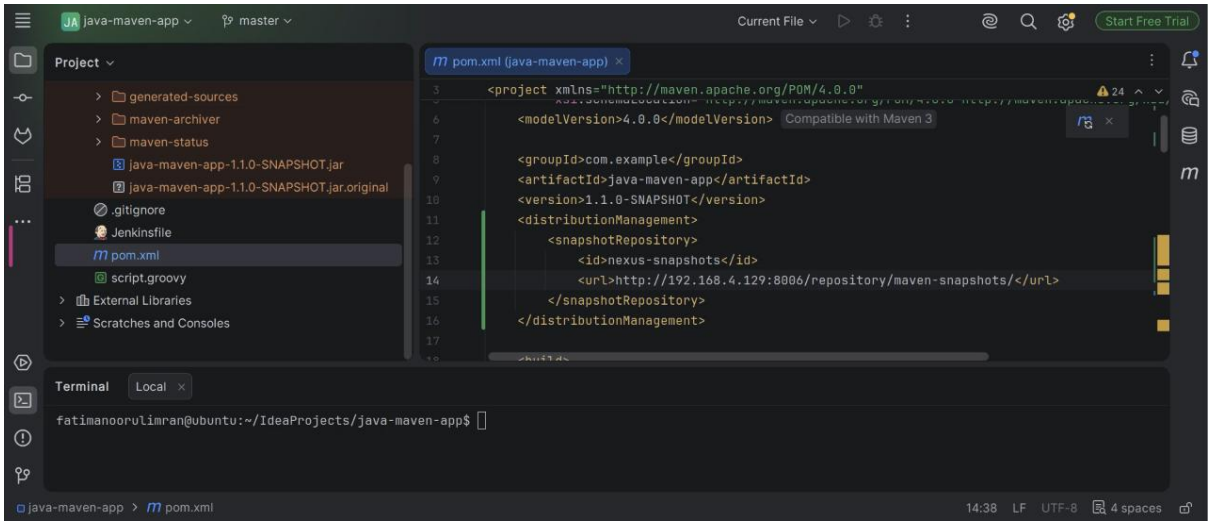


The my-app snapshot JAR and Maven metadata under com/example/my-app/1.0-SNAPSHOT in maven-snapshots.

Task 4: Publish the Maven Application to Nexus



The java-maven-app project opened in IntelliJ IDEA.



The distributionManagement block with the nexus-snapshots repository added to pom.xml.

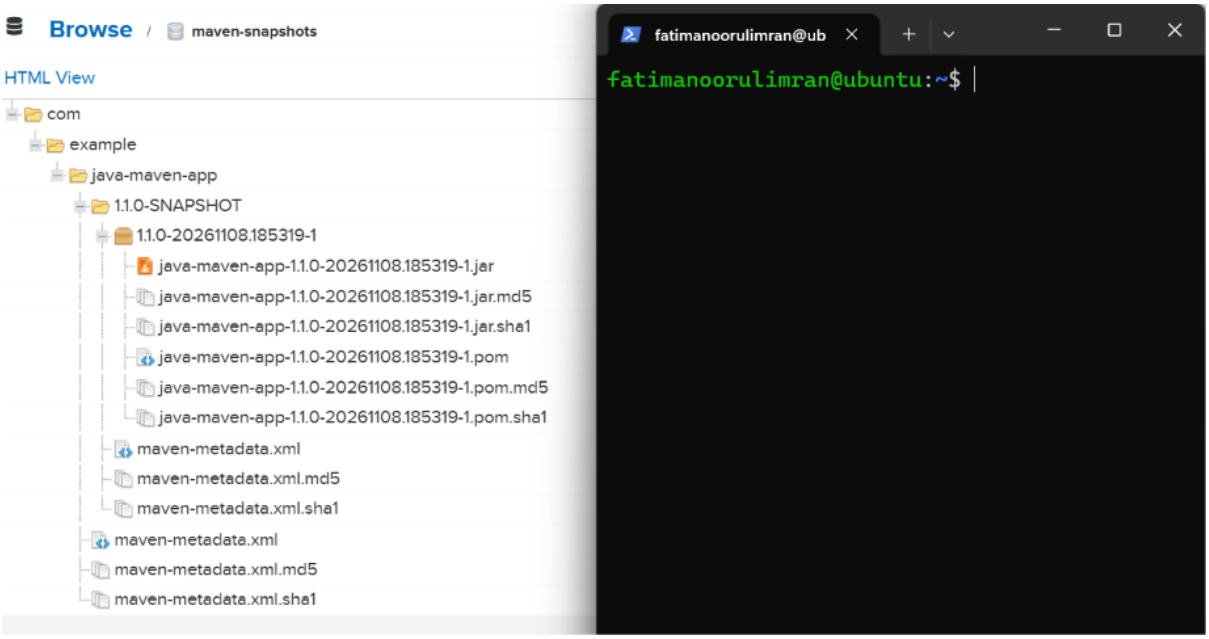
```
Terminal   Local

fatimanoorulimran@ubuntu:~/IdeaProjects/java-maven-app$ mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------------< com.example:java-maven-app >---------------------
[INFO] Building java-maven-app 1.1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.6.0:compile (default-compile) @ java-maven-app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /config/IdeaProjects/java-maven-app/src/test/resources
[INFO]
[INFO] --- compiler:3.6.0:testCompile (default-testCompile) @ java-maven-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:2.17:test (default-test) @ java-maven-app ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.1.2:jar (default-jar) @ java-maven-app ---
[INFO] Building jar: /config/IdeaProjects/java-maven-app/target/java-maven-app-1.1.0-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:2.3.5.RELEASE:repackage (default) @ java-maven-app ---
[INFO] Replacing main artifact with repackaged archive
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.044 s
[INFO] Finished at: 2026-11-08T17:01:43+05:00
```

mvn package completed with BUILD SUCCESS and built java-maven-app-1.1.0-SNAPSHOT.jar.

```
fatimanoorulimran@ubuntu:~/IdeaProjects/java-maven-app$ mvn deploy
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------------< com.example:java-maven-app >---------------------
[INFO] Building java-maven-app 1.1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.6.0:compile (default-compile) @ java-maven-app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /config/IdeaProjects/java-maven-app/src/test/resources
[INFO]
[INFO] --- compiler:3.6.0:testCompile (default-testCompile) @ java-maven-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:2.17:test (default-test) @ java-maven-app ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.1.2:jar (default-jar) @ java-maven-app ---
[INFO]
[INFO] --- spring-boot:2.3.5.RELEASE:repackage (default) @ java-maven-app ---
[INFO] Replacing main artifact with repackaged archive
[INFO]
[INFO] --- install:3.1.2:install (default-install) @ java-maven-app ---
[INFO] Installing /config/IdeaProjects/java-maven-app/pom.xml to /config/.m2/repository/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-SNAPSHOT.pom
[INFO] Installing /config/IdeaProjects/java-maven-app/target/java-maven-app-1.1.0-SNAPSHOT.jar to /config/.m2/repository/com/example/java-maven-app/1.1.0-SNAPSHOT/java-mav
en-app-1.1.0-SNAPSHOT.jar
[INFO]
[INFO] --- deploy:3.1.2:deploy (default-deploy) @ java-maven-app ---
Downloading from nexus-snapshots: http://192.168.4.129:8006/repository/maven-snapshots/com/example/java-maven-app/1.1.0-SNAPSHOT/maven-metadata.xml
Downloaded from nexus-snapshots: http://192.168.4.129:8006/repository/maven-snapshots/com/example/java-maven-app/1.1.0-SNAPSHOT/maven-metadata.xml (775 B at 983 B/s)
Uploading to nexus-snapshots: http://192.168.4.129:8006/repository/maven-snapshots/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-20261108.190159-2.pom
Uploading to nexus-snapshots: http://192.168.4.129:8006/repository/maven-snapshots/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-20261108.190159-2.jar
Uploaded to nexus-snapshots: http://192.168.4.129:8006/repository/maven-snapshots/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-20261108.190159-2.pom (2.9
 kB at 10 kB/s)
Uploaded to nexus-snapshots: http://192.168.4.129:8006/repository/maven-snapshots/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-20261108.190159-2.jar (39
MB at 18 MB/s)
Downloading from nexus-snapshots: http://192.168.4.129:8006/repository/maven-snapshots/com/example/java-maven-app/maven-metadata.xml
Downloaded from nexus-snapshots: http://192.168.4.129:8006/repository/maven-snapshots/com/example/java-maven-app/maven-metadata.xml (285 B at 2.5 kB/s)
Uploading to nexus-snapshots: http://192.168.4.129:8006/repository/maven-snapshots/com/example/java-maven-app/1.1.0-SNAPSHOT/maven-metadata.xml
Uploaded to nexus-snapshots: http://192.168.4.129:8006/repository/maven-snapshots/com/example/java-maven-app/1.1.0-SNAPSHOT/maven-metadata.xml (775 B at 3.8 kB/s)
Uploading to nexus-snapshots: http://192.168.4.129:8006/repository/maven-snapshots/com/example/java-maven-app/maven-metadata.xml
Uploaded to nexus-snapshots: http://192.168.4.129:8006/repository/maven-snapshots/com/example/java-maven-app/maven-metadata.xml (285 B at 1.8 kB/s)
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  11.589 s
```

mvn deploy uploaded the JAR, POM and metadata to nexus-snapshots and finished with BUILD SUCCESS.



The java-maven-app 1.1.0-SNAPSHOT JAR, POM and Maven metadata visible in maven-snapshots.

Task 5: Inspect Nexus Repositories and Components Using the REST API

```
fatimanoorulimran@ubuntu:~$ curl -u nexus -X GET "http://192.168.4.129:8006/service/rest/v1/repositories"
Enter host password for user 'nexus':
[ {
  "name" : "maven-snapshots",
  "format" : "maven2",
  "type" : "hosted",
  "url" : "http://192.168.4.129:8006/repository/maven-snapshots",
  "online" : true,
  "size" : 0,
  "attributes" : { }
} ]fatimanoorulimran@ubuntu:~$
```

Repositories visible to the nexus user: only maven-snapshots.

```
fatimanoorulimran@ubuntu:~$ curl -u admin -X GET "http://192.168.4.129:8006/service/rest/v1/repositories"
Enter host password for user 'admin':
[ {
  "name" : "maven-releases",
  "format" : "maven2",
  "type" : "hosted",
  "url" : "http://192.168.4.129:8006/repository/maven-releases",
  "online" : true,
  "size" : 0,
  "attributes" : { }
}, {
  "name" : "maven-snapshots",
  "format" : "maven2",
  "type" : "hosted",
  "url" : "http://192.168.4.129:8006/repository/maven-snapshots",
  "online" : true,
  "size" : 0,
  "attributes" : { }
}, {
  "name" : "maven-central",
  "format" : "maven2",
  "type" : "proxy",
  "url" : "http://192.168.4.129:8006/repository/maven-central",
  "online" : true,
  "size" : 0,
  "attributes" : {
    "proxy" : {
      "remoteUrl" : "https://repo1.maven.org/maven2/"
    }
  }
}, {
  "name" : "maven-public",
  "format" : "maven2",
  "type" : "group",
  "url" : "http://192.168.4.129:8006/repository/maven-public",
  "online" : true,
  "size" : 0,
  "attributes" : { }
}, {
  "name" : "nuget-hosted",
  "format" : "nuget",
  "type" : "hosted",
  "url" : "http://192.168.4.129:8006/repository/nuget-hosted",
  "online" : true,
  "size" : 0,
  "attributes" : { }
}, {
  "name" : "nuget.org-proxy",
  "format" : "nuget",
```

Repositories visible to the admin user: the full repository list.

```
fatimanoorulimran@ubuntu:~$ curl -u nexus -X GET "http://192.168.4.129:8006/service/rest/v1/components?repository=maven-snapshots"
Enter host password for user 'nexus':
{
  "items" : [ {
    "id" : "bWF2ZW4tc25hcHNob3RzOjRmMWJiY2Rk",
    "repository" : "maven-snapshots",
    "format" : "maven2",
    "group" : "com.example",
    "name" : "my-app",
    "version" : "1.0-20261108.114454-1",
    "assets" : [ {
      "downloadUrl" : "http://192.168.4.129:8006/repository/maven-snapshots/com/example/my-app/1.0-SNAPSHOT/my-app-1.0-20261108.114454-1.jar",
      "path" : "/com/example/my-app/1.0-SNAPSHOT/my-app-1.0-20261108.114454-1.jar",
      "id" : "bWF2ZW4tc25hcHNob3RzOmRhYTY2ZDJl",
      "repository" : "maven-snapshots",
      "format" : "maven2",
      "checksum" : {
        "md5" : "2bbce31b41da55e2c7bc203f8b56d2e0",
        "sha1" : "6e8e099359a0aab9e76f175a61933cf60c8792ac",
        "sha256" : "9abbb2b2739e435f83ea836f7597ea8584457ef77bbed12dc450bce6558d54f6",
        "sha512" : "21530f3eec96390a39f3b951f3fd4d1b3400851cad4b59d1b11ac955a76ae2a106a0211a56a58ba99780f817d800d8dd0749235a0eea468d86b10acc5bfb
6dd6"
      },
      "contentType" : "application/java-archive",
      "lastModified" : "2026-11-08T11:45:02.192+00:00",
      "lastDownloaded" : null,
      "uploader" : null,
      "uploaderIp" : null,
      "fileSize" : 39682472,
      "blobCreated" : "2026-11-08T11:45:02.200+00:00",
      "blobStoreName" : "default",
      "blobUpdated" : "2026-11-08T11:45:02.192+00:00",
      "blobRef" : "default@3c38e6d7-5012-4d2f-a590-b6e85b99cf27@2026-11-08T11:45",
      "lastVerified" : null,
      "registryUrl" : null,
      "maven2" : {
```

Components in maven-snapshots listed through the REST API as the nexus user.

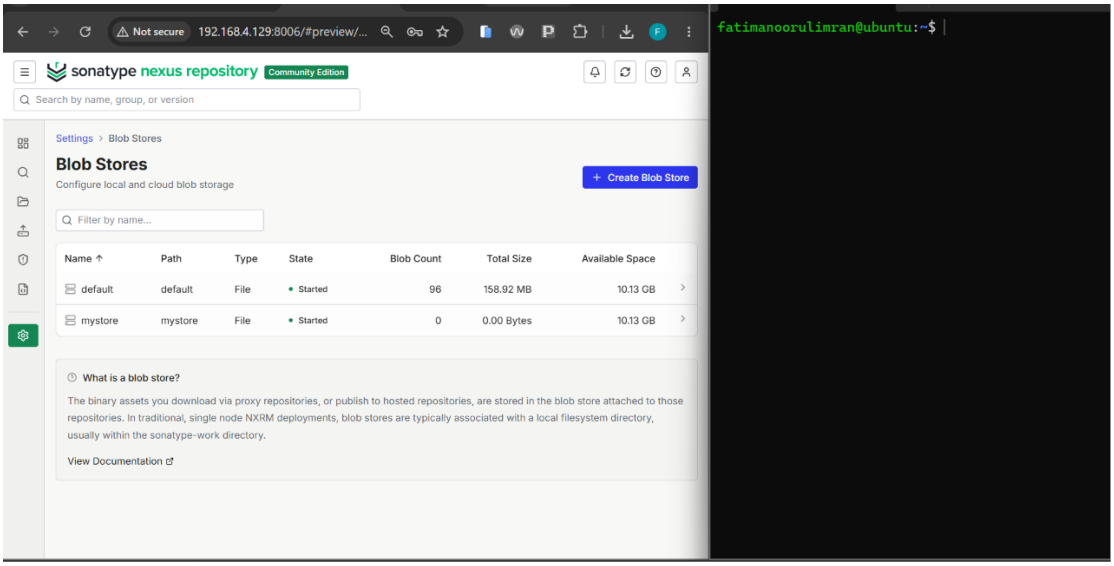
```
fatimanoorulimran@ubuntu:~$ curl -u nexus -X GET "http://192.168.4.129:8006/service/rest/v1/components/bWF2ZW4tc25hcHNob3RzOjRmMWJiY2Rk"
Enter host password for user 'nexus':
{
  "id" : "bWF2ZW4tc25hcHNob3RzOjRmMWJiY2Rk",
  "repository" : "maven-snapshots",
  "format" : "maven2",
  "group" : "com.example",
  "name" : "my-app",
  "version" : "1.0-20261108.114454-1",
  "assets" : [ {
    "downloadUrl" : "http://192.168.4.129:8006/repository/maven-snapshots/com/example/my-app/1.0-SNAPSHOT/my-app-1.0-20261108.114454-1.jar",
    "path" : "/com/example/my-app/1.0-SNAPSHOT/my-app-1.0-20261108.114454-1.jar",
    "id" : "bWF2ZW4tc25hcHNob3RzOmRhYTY2ZDJl",
    "repository" : "maven-snapshots",
    "format" : "maven2",
    "checksum" : {
      "md5" : "2bbce31b41da55e2c7bc203f8b56d2e0",
      "sha1" : "6e8e099359a0aab9e76f175a61933cf60c8792ac",
      "sha256" : "9abbb2b2739e435f83ea836f7597ea8584457ef77bbed12dc450bce6558d54f6",
      "sha512" : "21530f3eec96390a39f3b951f3fd4d1b3400851cad4b59d1b11ac955a76ae2a106a0211a56a58ba99780f817d800d8dd0749235a0eea468d86b10acc5bfb6d
d6"
    },
    "contentType" : "application/java-archive",
    "lastModified" : "2026-11-08T11:45:02.192+00:00",
    "lastDownloaded" : null,
    "uploader" : null,
    "uploaderIp" : null,
    "fileSize" : 39682472,
    "blobCreated" : "2026-11-08T11:45:02.200+00:00",
    "blobStoreName" : "default",
    "blobUpdated" : "2026-11-08T11:45:02.192+00:00",
    "blobRef" : "default@3c38e6d7-5012-4d2f-a590-b6e85b99cf27@2026-11-08T11:45",
    "lastVerified" : null,
    "registryUrl" : null,
    "maven2" : {
      "extension" : "jar",
```


Details of one component retrieved by its ID.

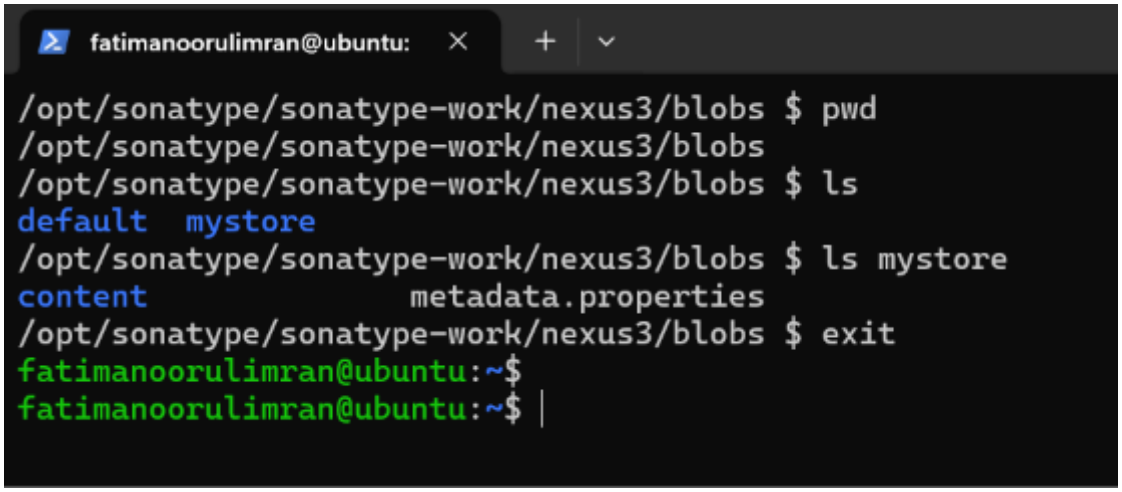
Task 6: Create a Nexus Blob Store and Docker Hosted Repository

```
fatimanoorulimran@ubuntu:~$ docker exec -it nexus /bin/sh
/opt/sonatype $ pwd
/opt/sonatype
/opt/sonatype $ cd sonatype-work/nexus3
/opt/sonatype/sonatype-work/nexus3 $ cd blobs
/opt/sonatype/sonatype-work/nexus3/blobs $ ls
default
/opt/sonatype/sonatype-work/nexus3/blobs $ ls default/content
2026        directpath  tmp
/opt/sonatype/sonatype-work/nexus3/blobs $
```

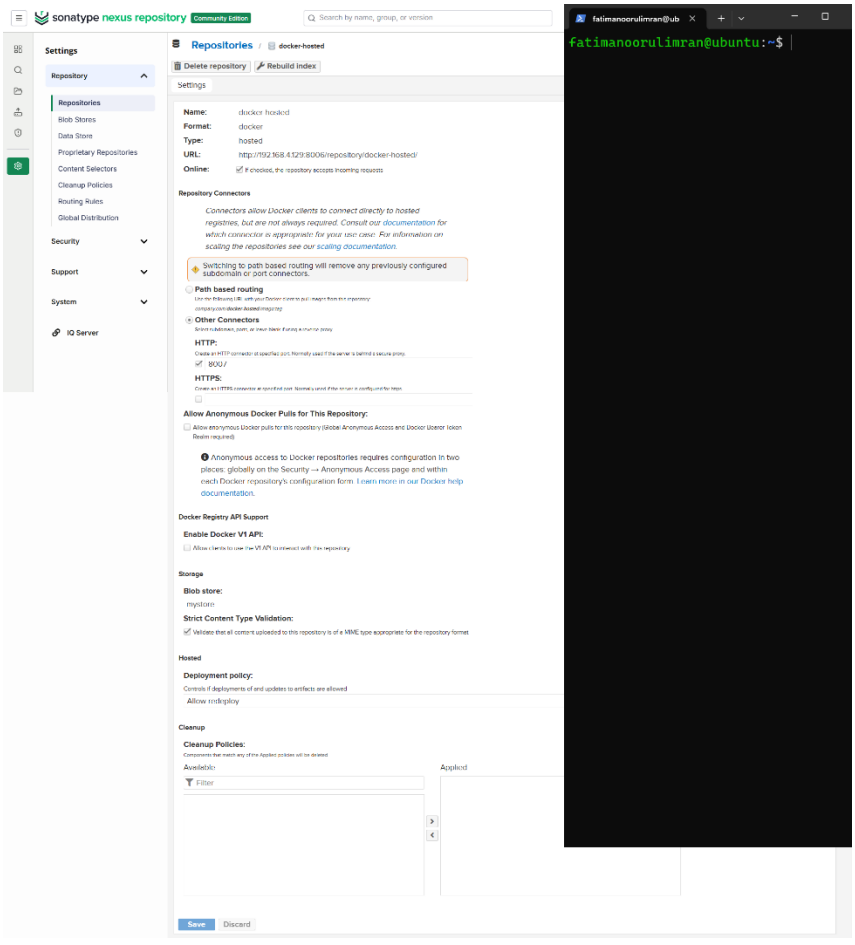
Contents of the default blob store directory inside the Nexus container.



Blob Stores page showing default and mystore, both File type and Started.

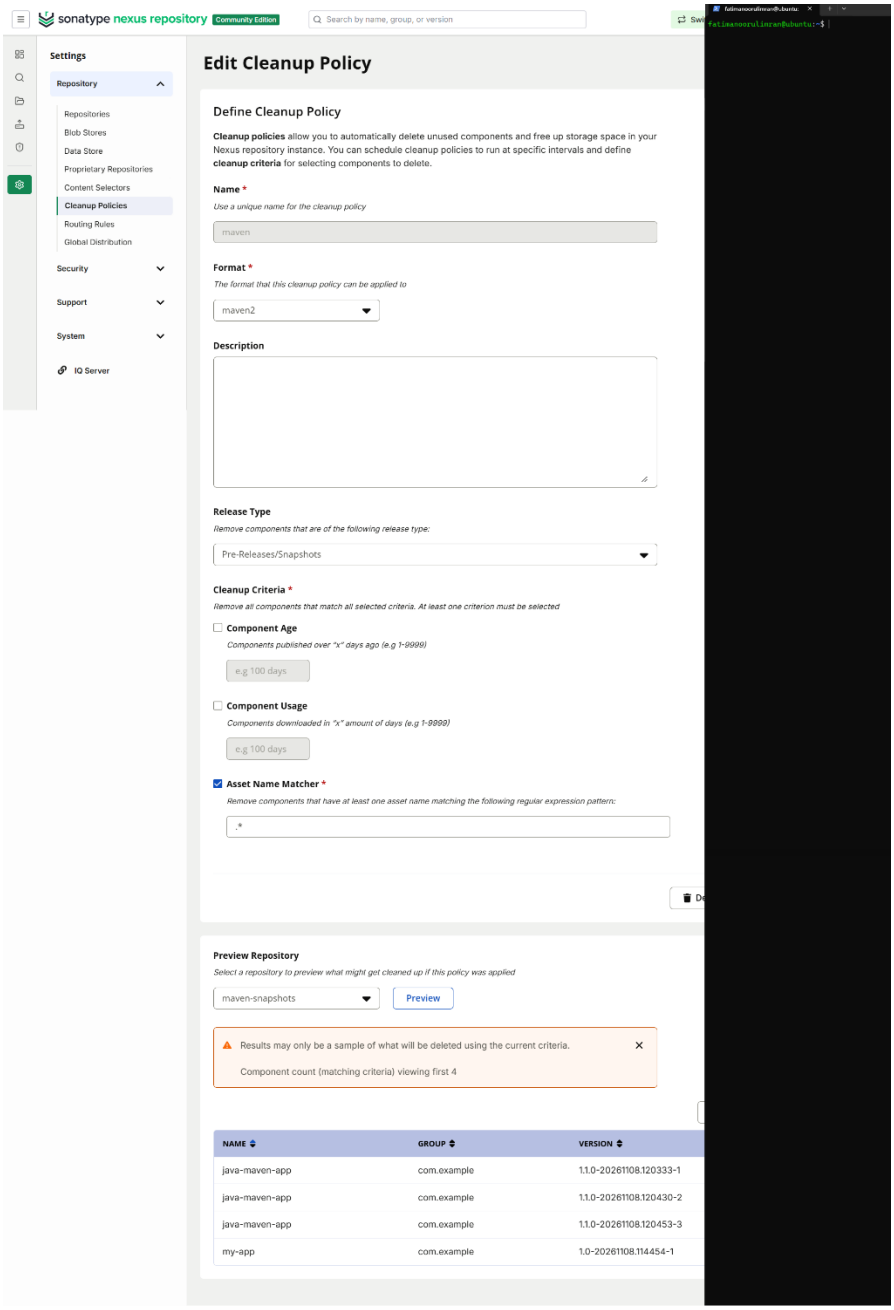


The blobs directory inside the container now contains default and mystore.

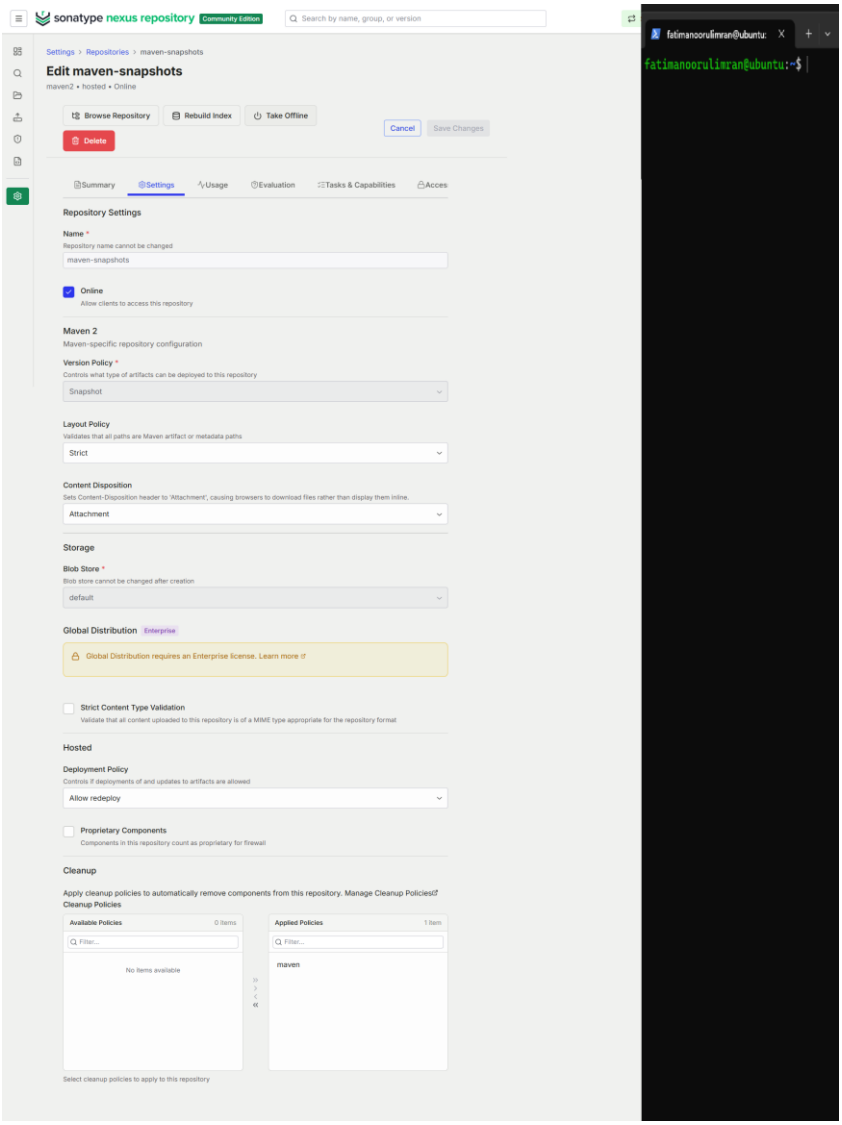


The docker-hosted repository using the mystore blob store with HTTP connector 8007.

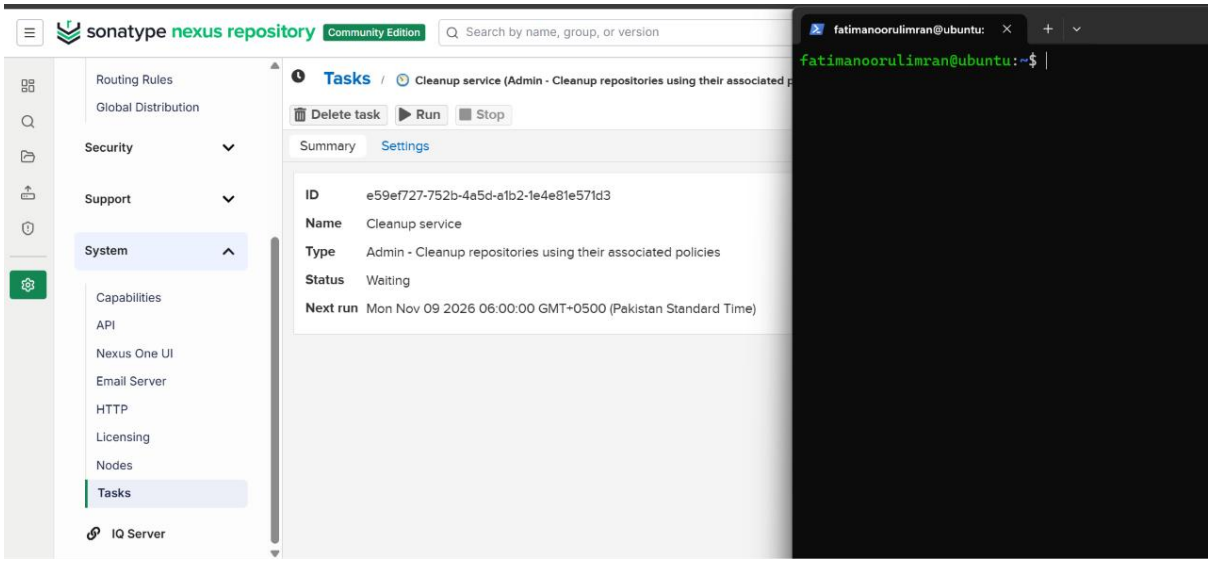
Task 7: Create and Run a Maven Snapshot Cleanup Policy



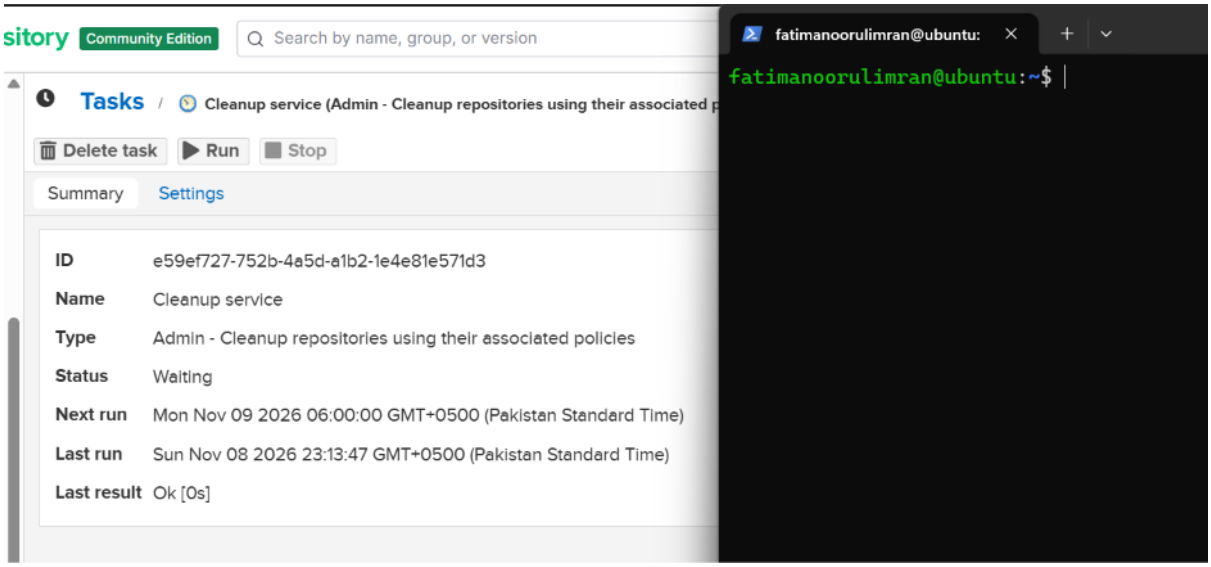
The maven cleanup policy (maven2, Pre-Releases/Snapshots, .* matcher) with the maven-snapshots preview.



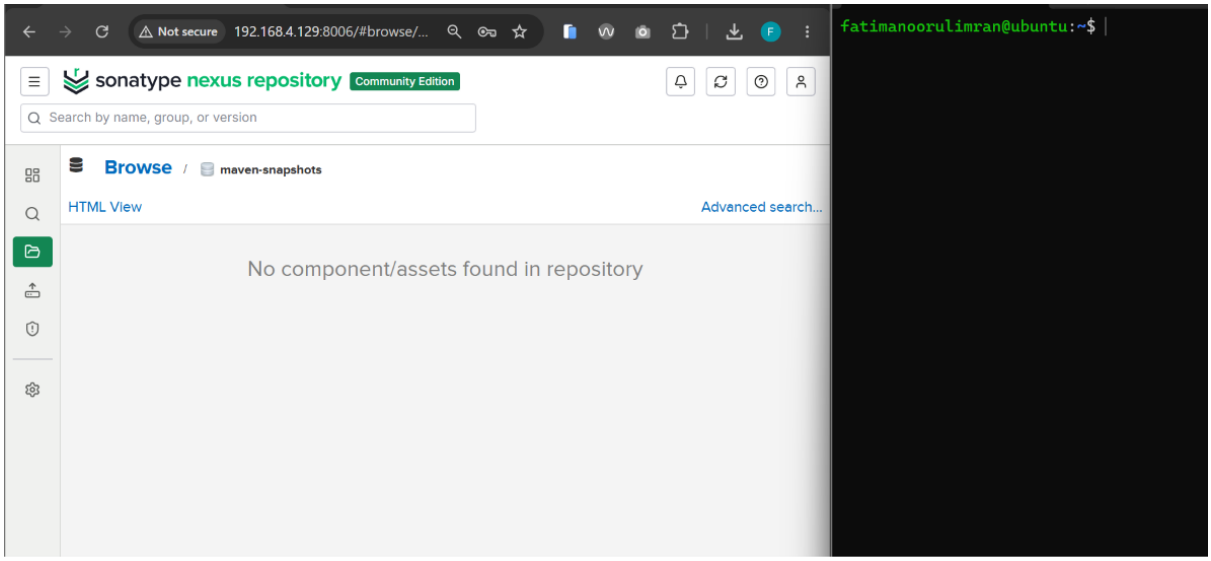
The maven policy applied to the maven-snapshots repository.



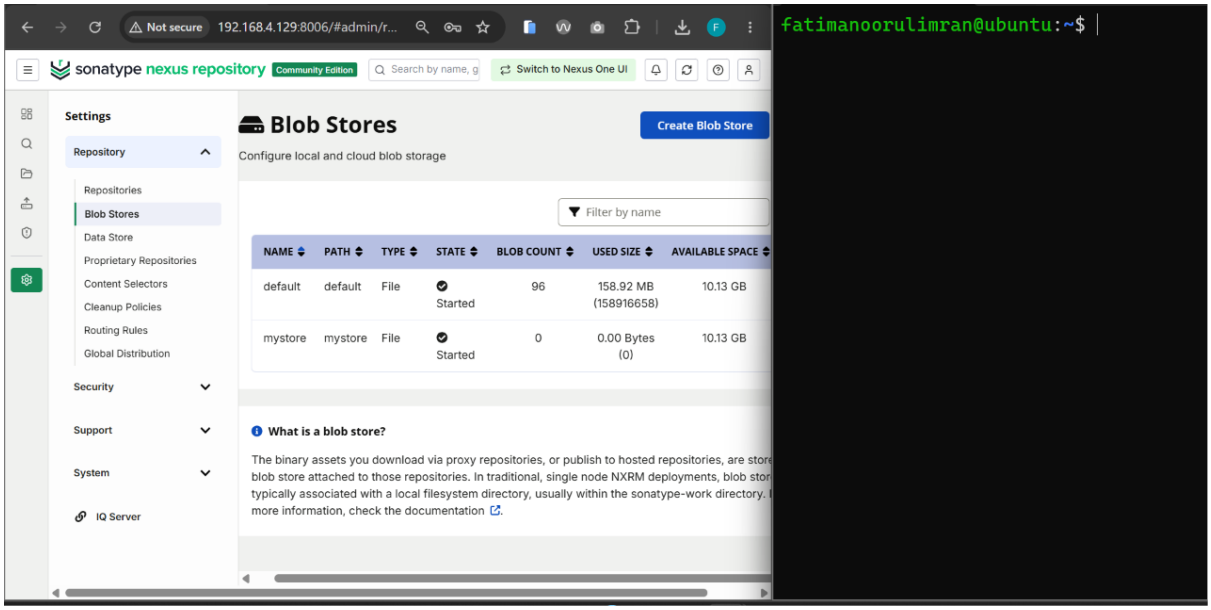
Cleanup service task in Waiting state with the next run scheduled for 06:00.



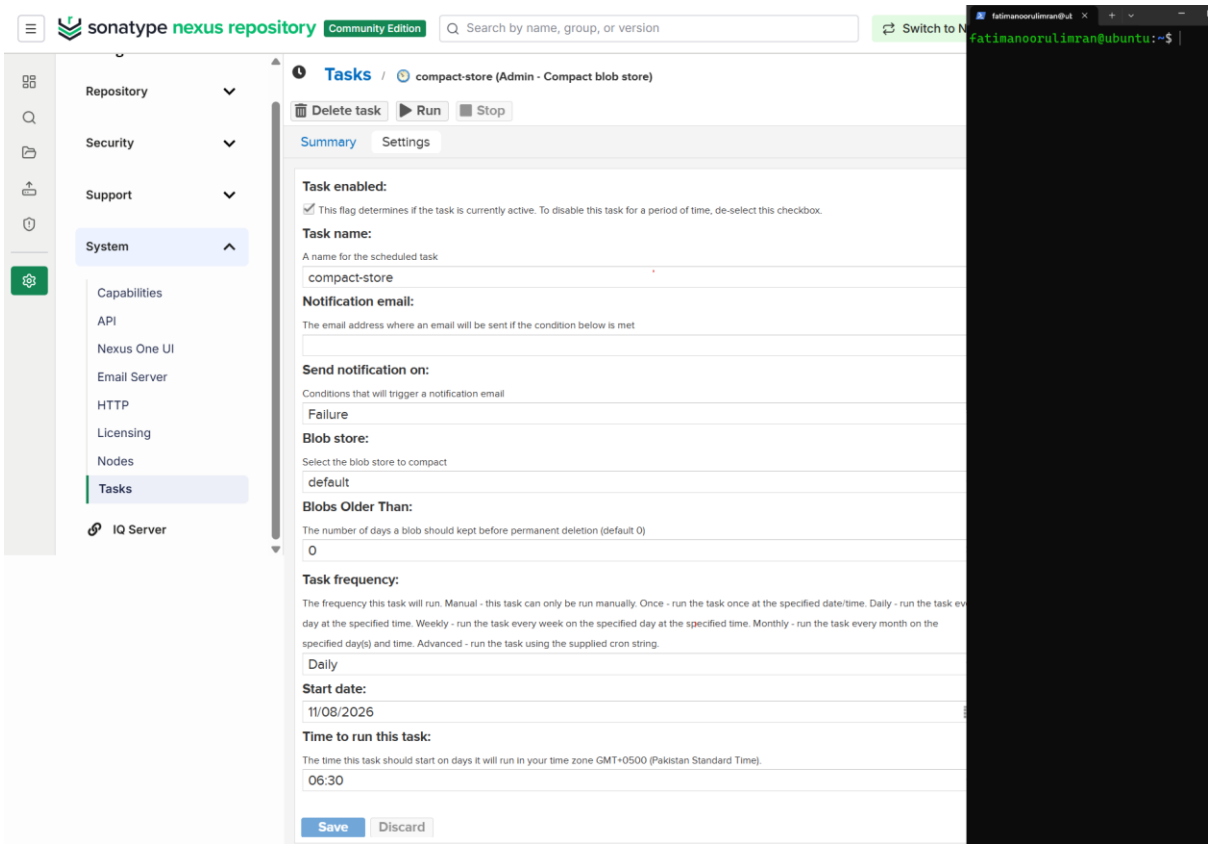
Cleanup service after the manual run, with Last result: Ok.



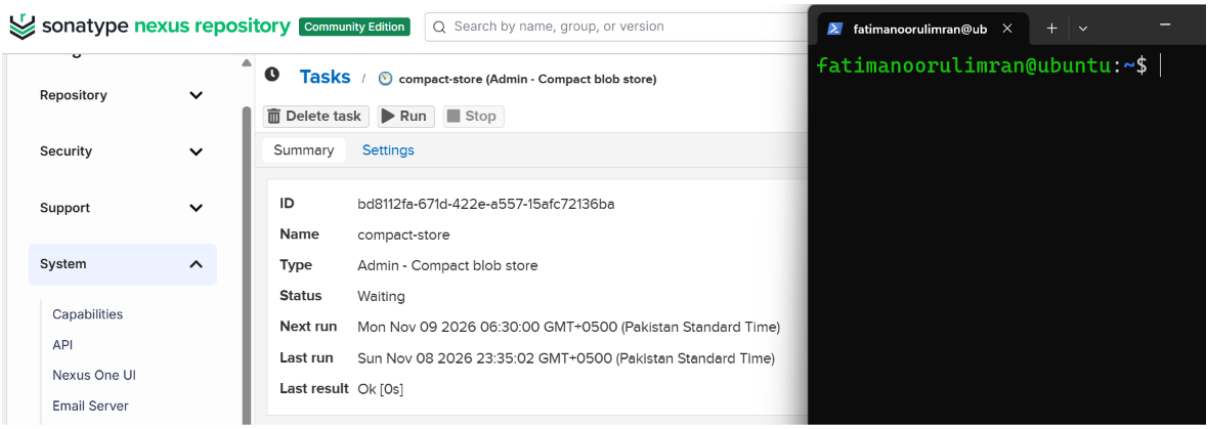
maven-snapshots is empty after the cleanup: No component/assets found in repository.



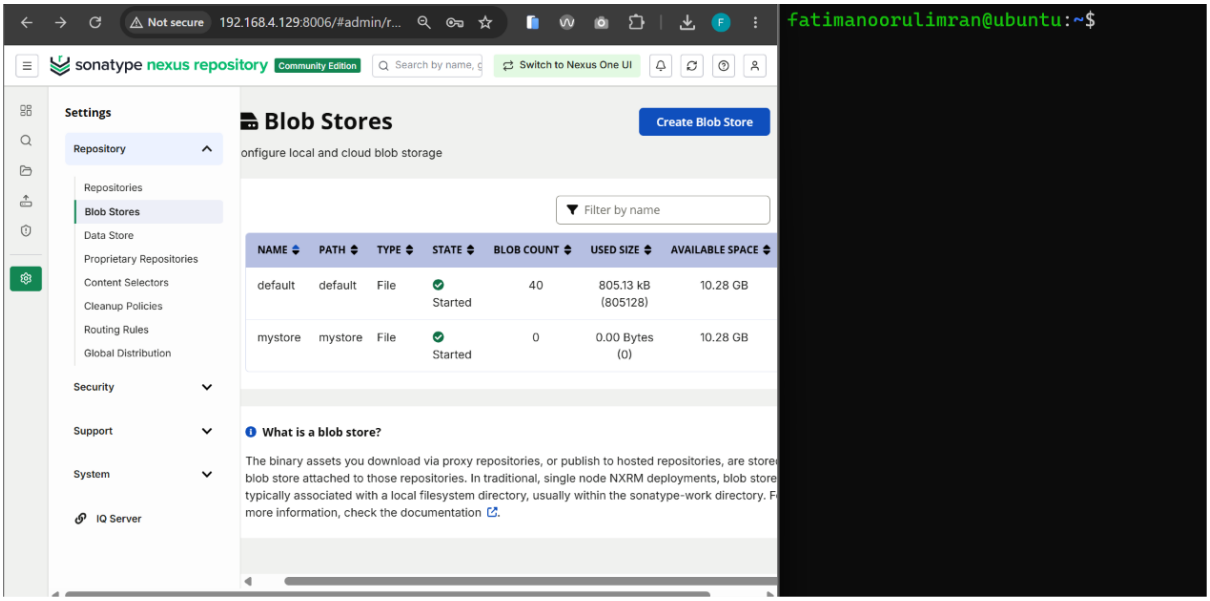
Default blob store before compaction: 96 blobs, 158.92 MB used.



The compact-store task configured for the default blob store, 0 days, daily at 06:30.



compact-store after the manual run, with Last result: Ok.



Default blob store after compaction: 40 blobs, 805.13 kB used, available space increased.